

Household Financial Resilience Index

A Mathematical Framework for Measuring Household Financial Resilience

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Research Question

Can a transparent mathematical index combining savings, debt, emergency liquidity, and expenditure stability provide a meaningful measure of household financial resilience?

1 Abstract

Financial resilience is the ability of a household to absorb a financial shock while continuing to meet its essential financial obligations. Income alone does not fully describe this ability. Two households with similar incomes can have very different financial positions if one has liquid savings and manageable debt while the other has little liquidity, high repayment obligations, or unstable expenditure.

This paper develops the Household Financial Resilience Index (HFRI), a transparent composite measure based on four dimensions: Savings Strength, Debt Sustainability, Emergency Coverage, and Expense Stability. Five underlying indicators are converted into bounded 0–100 scores and combined into a single index. The model was first tested using 1,000 simulated households observed for 12 months, producing 12,000 household-month observations. Phase I established the mathematical framework, calibrated the scoring functions, examined internal behaviour, and tested weighting sensitivity. Phase II introduced an external benchmark based on the Office for National Statistics (ONS) financial-resilience measure and subjected the simulated households to two controlled financial shocks.

The final HFRI scores ranged from 16.27 to 97.24, with a mean of 75.84 and median of 82.04. The index moved in the expected direction relative to its underlying indicators. The lowest rank correlation under the broader tested weighting alternatives was 0.9707. In Phase II, 75.9% of simulated households had enough liquid savings to cover a 25% income reduction lasting three months, and HFRI had a Pearson correlation of 0.8729 with the resulting binary benchmark. Shock survival also increased sharply with HFRI, with the main transition occurring around 65–70.

The findings support HFRI as a coherent measurement framework within the simulated environment. They do not establish real-world predictive validity. The interpretation bands are therefore provisional and should ultimately be tested against observed household-level data.

2 Introduction and Research Question

Household financial resilience is difficult to represent with one financial variable. Income measures the resources entering a household, but not how much remains after expenditure. Savings measure accumulated resources, but do not show the debt burden or how predictable expenditure is. Debt measures financial obligations, but the size of debt alone does not show whether current repayments are manageable. Average expenditure also does not show whether spending is stable from month to month.

This creates a measurement problem: a household's ability to withstand a shock depends on several financial conditions at the same time. The HFRI therefore treats financial resilience as a multidimensional concept:

$$\text{Financial Resilience} = f(\text{Savings, Debt, Emergency Resources, Expense Stability}).$$

For this research, financial resilience is defined as the capacity of a household to absorb and withstand financial shocks while continuing to meet its essential financial obligations.

Research Question

Can a transparent mathematical index combining savings, debt, emergency liquidity, and expenditure stability provide a meaningful measure of household financial resilience?

The objective is not to claim that HFRI is already a validated predictor of financial distress. The first objective is to construct a measure whose assumptions are visible, mathematically reproducible, and economically interpretable. The second is to test whether it behaves logically under controlled financial conditions. Phase II extends this by comparing the index with an established external framework and examining its response to defined shocks.

3 Structure of the HFRI

The HFRI contains four dimensions:

1. **Savings Strength:** the household's ability to generate a financial surplus.
2. **Debt Sustainability:** the size of debt and the burden of servicing it.
3. **Emergency Coverage:** liquid savings relative to essential expenditure.
4. **Expense Stability:** the predictability of household expenditure.

The five underlying indicators are:

$$SR = \frac{Y - C}{Y}, \quad DTI = \frac{D_b}{Y}, \quad DSTI = \frac{DS}{Y},$$

$$EC = \frac{L}{C_{\text{essential},m}}, \quad CV = \frac{\sigma_C}{\bar{C}}.$$

Here:

- Y = annual disposable household income;
- C = annual household expenditure;
- D_b = outstanding household debt;
- DS = annual required debt-service payments;
- L = liquid savings;
- $C_{\text{essential},m}$ = average monthly essential expenditure;
- σ_C = standard deviation of monthly expenditure;
- $\bar{C}$ = mean monthly expenditure.

Each indicator is transformed into a score between 0 and 100, where a higher score always represents greater resilience.

The key distinction is between flow and stock variables. Savings Strength measures the household’s current capacity to generate a surplus, while Emergency Coverage measures the liquid resources already accumulated. Debt Sustainability captures both accumulated leverage and current repayment pressure. Expense Stability captures the predictability of financial requirements.

4 Mathematical Specification and Calibration

4.1 Savings Strength

Savings Strength begins with the savings rate:

$$SR = \frac{Y - C}{Y}$$

A positive value means income exceeds expenditure; a negative value means expenditure exceeds income. The personal saving rate is similarly defined as saving relative to disposable income in official national-account statistics [2, 3, 1].

The final scoring function is:

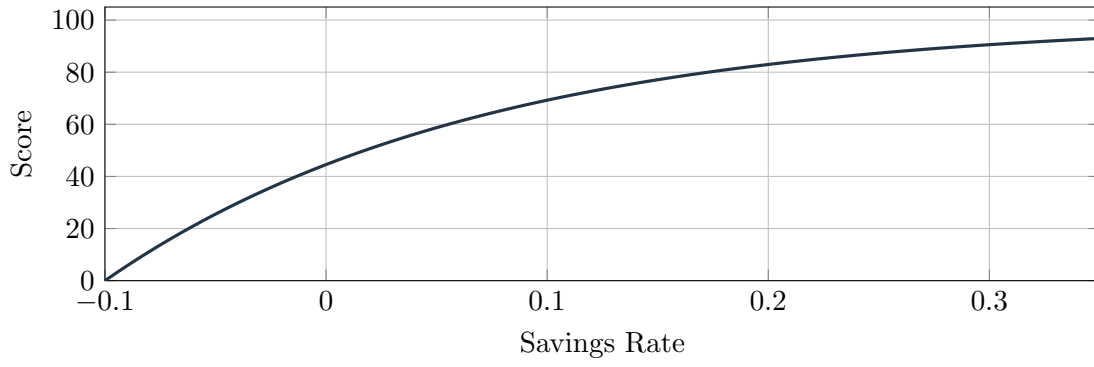
$$S(SR) = \begin{cases} 0, & SR \leq -0.10, \\ 100 (1 - e^{-5.89681(SR+0.10)}) , & SR > -0.10. \end{cases}$$

The function is increasing and concave. This means that moving from a weak savings position to a moderate one produces a larger increase in score than making an already high savings rate slightly higher.

The researcher-defined calibration points were:

Savings Rate	Score
-10%	0
-4%	25
0%	50
3%	55
7%	62
12%	70
17%	77
25%	88
30%	95

The calibration was informed by household-saving literature and practical savings-rate guidance. Vanguard suggests saving 12–15% of pay for retirement, while Fidelity uses 15% as a retirement-saving guideline [5, 6]. The 20% savings component in the 50/30/20 framework was also treated as a contextual budgeting reference rather than a resilience threshold [8]. BCG provides additional context on savings behaviour across Indian consumer groups [9].



[Open the Savings Strength Desmos model](#)

Figure 1. Savings Strength calibration function. The plotted curve uses the final HFRI equation; the link opens the corresponding Desmos model.

Takeaway: Savings Strength measures ongoing surplus generation, but it is deliberately not treated as a substitute for liquid emergency resources. A household can have a strong current savings rate and still have little accumulated liquidity.

4.2 Debt Sustainability

Debt Sustainability contains two separate underlying indicators because debt creates two different forms of financial pressure.

First:

$$DTI = \frac{D_b}{Y}$$

where D_b is outstanding household debt and Y is annual disposable household income.

DTI measures the **stock of accumulated debt** relative to the household's income base.

Second:

$$DSTI = \frac{DS}{Y}$$

where DS is annual required debt-service payments.

DSTI measures the **current recurring repayment burden**.

This distinction is important. A household can have substantial outstanding debt but relatively manageable repayments, while another can have a smaller debt stock but a high current repayment burden.

The final DTI scoring function is:

$$D_{DTI} = 100e^{-0.0573534(DTI)^{2.13509}}$$

The final DSTI function is:

$$D_{DSTI} = -521.45598(DSTI)^2 + 9.08245(DSTI) + 100.00523$$

The DSTI score is operationally bounded:

$$D_{DSTI, \text{operational}} = \min(100, \max(0, D_{DSTI}))$$

The final Debt Sustainability score is:

$$D = 0.40D_{DTI} + 0.60D_{DSTI}$$

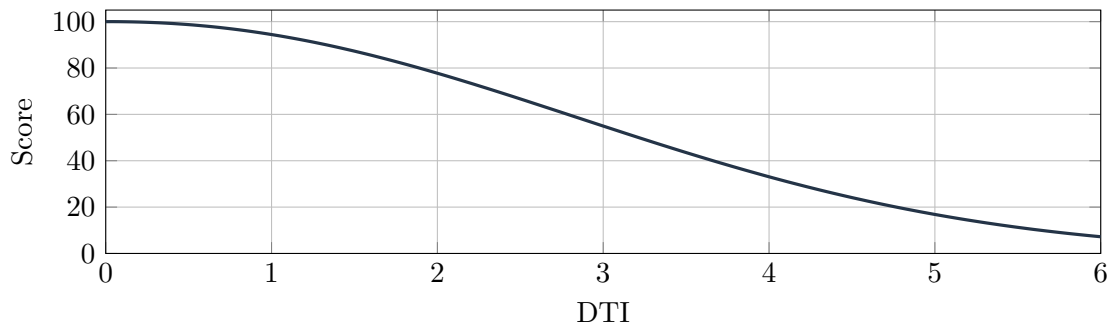
The 40:60 weighting gives greater importance to current repayment pressure while retaining accumulated leverage as a separate measure.

DTI Calibration

The DTI calibration points were:

DTI	DTI Score
0%	100
250%	65
300%	55
350%	45
450%	25
600%	5

The underlying literature provides useful landmarks but not a universal household-resilience threshold. Bank of Canada analysis identifies households at or above 350% debt-to-gross-income as highly indebted, while BIS/ECB work has used approximately 300% as a vulnerability reference point [10, 11]. UK mortgage research also provides evidence around high loan-to-income levels, although mortgage-origination rules are not directly equivalent to a household-level resilience measure [14, 12]. APRA's six-times-income threshold is similarly a lending-policy instrument rather than a universal resilience cut-off [15].



[Open the DTI Desmos model](#)

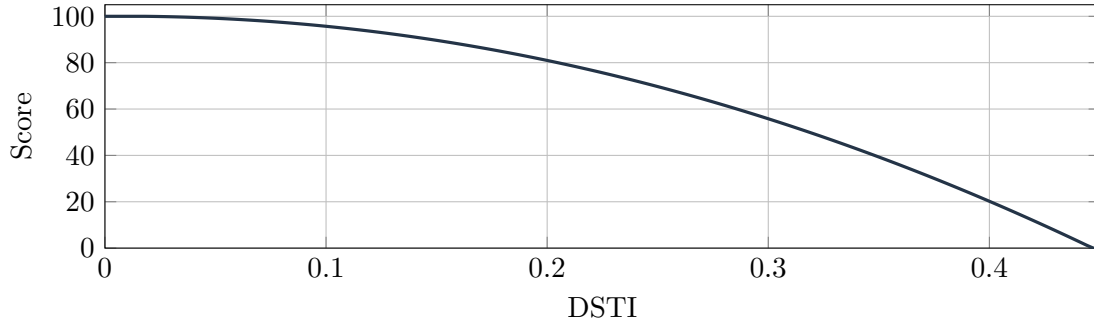
Figure 2. DTI-to-score calibration function.

DSTI Calibration

The DSTI calibration points were:

DSTI	DSTI Score
0%	100
25%	70
30%	55
35%	40
40%	20

These points were informed by evidence concerning high debt-service burdens. The Bank of England reports that repayment difficulties can rise sharply when mortgage debt-service ratios are above approximately 35–40%, while BIS household debt-service-ratio data are used as indicators of financial vulnerability [12, 13, 16].



[Open the DSTI Desmos model](#)

Figure 3. DSTI-to-score calibration function. The operational score is clipped to the 0–100 interval.

The selected DTI function achieved:

$$R^2 = 0.9980,$$

while the selected DSTI function achieved:

$$R^2 = 0.9997.$$

The functions were not selected by R^2 alone. Alternative linear, quadratic, cubic, exponential, logistic and power-exponential specifications were examined. The final choices were made using goodness of fit together with monotonicity, boundedness, economic interpretation and extrapolation behaviour.

Takeaway: DTI and DSTI overlap but are not redundant. In the simulation, their raw correlation was 0.7659, while the correlation between their transformed scores was 0.6536. The final 40:60 weighting therefore reflects a conceptual preference for current repayment pressure rather than a statistically estimated “optimal” weight.

4.3 Emergency Coverage

Emergency Coverage measures the amount of essential expenditure that can be financed from immediately available liquid savings:

$$EC = \frac{L}{C_{\text{essential},m}}$$

Thus, $EC = 3$ means that liquid savings are equivalent to three months of essential expenditure.

The use of liquid resources rather than total wealth is important. Illiquid assets may contribute to household wealth without being immediately available to finance a temporary shock.

The scoring function is:

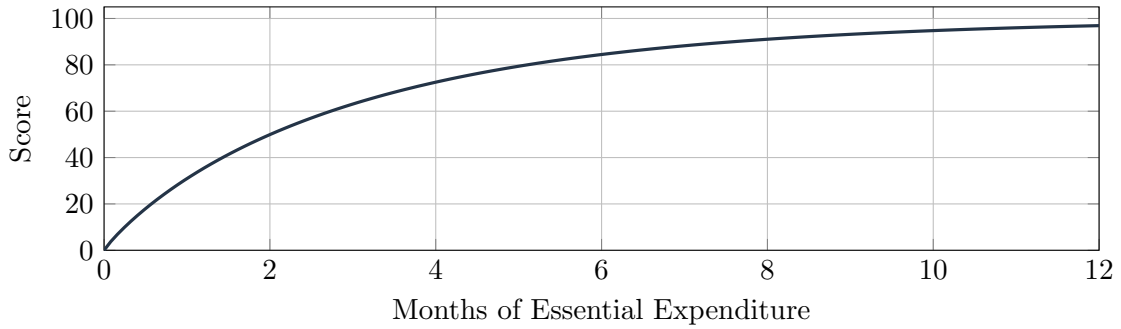
$$E(EC) = 100 \left(1 - e^{-0.37028EC^{0.901391}} \right)$$

The calibration points were:

Emergency Coverage	Score
0 months	0
0.6 months	20
1 month	30
3 months	65
6 months	85
9 months	92
12 months	96

The first six months are particularly relevant because emergency-savings research commonly examines three- to six-month buffers. Federal Reserve research finds that only around 40% of U.S. families had liquid savings equivalent to at least three months of expenses in the 2016 SCF, while fewer than 30% had six months [18]. Later Federal Reserve research also identifies three months as a common emergency-fund benchmark [19, 17]. OECD work provides related financial-security measures [20]. JPMorgan Chase research similarly finds that liquid assets are important for absorbing income and expenditure volatility [21, 22].

The 9- and 12-month points are researcher-defined extensions rather than literature-established thresholds.



[Open the Emergency Coverage Desmos model](#)

Figure 4. *Emergency Coverage-to-score calibration function.*

The selected function achieved:

$$R^2 = 0.9991.$$

It is increasing and concave:

$$\frac{dE}{dEC} > 0, \quad \frac{d^2E}{dEC^2} < 0.$$

This creates diminishing marginal gains: moving from almost no liquidity to several months of coverage produces a larger increase in resilience than adding further months to an already large buffer.

Takeaway: Emergency Coverage is the clearest direct measure in the HFRI of immediate liquidity-based shock absorption. It is deliberately separated from Savings Strength because a household can have a strong current savings rate without having accumulated a substantial emergency buffer.

4.4 Expense Stability

Expense Stability measures the predictability of household expenditure using the coefficient of variation:

$$CV = \frac{\sigma_C}{\bar{C}}$$

where σ_C is the standard deviation of monthly expenditure and $\bar{C}$ is mean monthly expenditure.

A lower CV represents greater stability.

The final scoring function is:

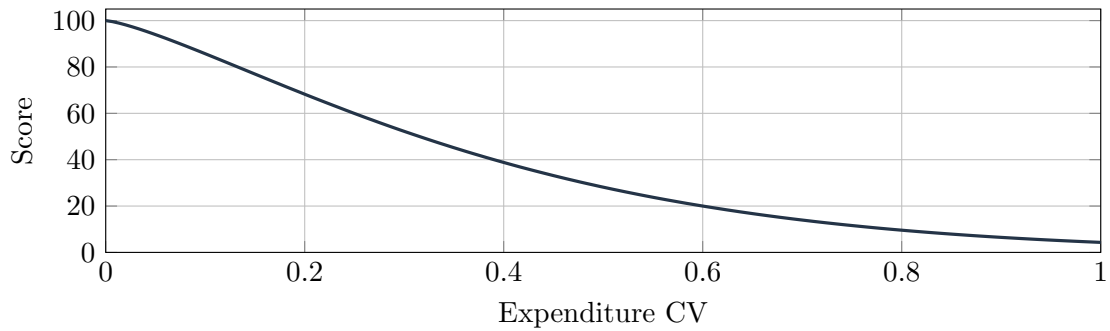
$$X(CV) = 100e^{-3.13957CV^{1.30837}}$$

The calibration points were:

CV	Expense Stability Score
0.00	100
0.05	92
0.10	83
0.20	68
0.30	53
0.33	50
0.40	40
0.50	28
0.60	19
0.80	8
1.00	3

The principal empirical calibration landmark was approximately $CV = 0.33$, based on JPMorgan Chase Institute evidence on spending volatility [22, 23]. JPMorgan Chase research also found that expenses changed by approximately 29% month-to-month for median-income households [24]. Other research associates greater consumption volatility with financial constraints, including evidence from Fisher and Hardy and Indian evidence from Chakrabarty and Swathysree [25, 26].

However, expenditure volatility is not equivalent to financial distress. A financially secure household may also make large discretionary or lumpy purchases. Expense Stability is therefore treated as the most exploratory HFRI dimension.



[Open the Expense Stability Desmos model](#)

Figure 5. Expense Stability calibration function.

The selected function achieved:

$$R^2 = 0.9980.$$

A cubic produced a slightly higher numerical fit in the original functional-form testing, but was rejected because polynomial extrapolation can create economically implausible behaviour outside the calibration range.

Takeaway: Expense Stability adds information that average expenditure alone cannot provide, but

its interpretation is deliberately narrower than that of savings or liquidity. It measures expenditure predictability, not overall financial health.

4.5 Calibration Method

The scoring functions were not selected only by maximizing R^2 . For each indicator, the process was:

1. define the underlying variable;
2. establish the expected economic direction;
3. identify literature-informed ranges and landmarks;
4. assign explicit researcher-defined score anchors;
5. test alternative mathematical forms;
6. check monotonicity, bounds, extrapolation and marginal behaviour; and
7. select the function using goodness of fit together with economic and mathematical logic.

The final calibration fits were:

Dimension	Calibration Points	R^2
Savings Strength	9	0.9876
DTI component	6	0.9980
DSTI component	5	0.9997
Emergency Coverage	7	0.9991
Expense Stability	11	0.9980

These R^2 values describe fit to researcher-defined calibration points. They are not evidence that the functions have been empirically validated.

Takeaway: The mathematical calibration establishes a reproducible transformation from each financial indicator to a common 0–100 scale. It does not establish that any particular score corresponds to a measured probability of financial distress.

5 Final Index and Weighting

The baseline HFRI is:

$$HFRI = 0.25S + 0.25D + 0.25E + 0.25X$$

where:

$$D = 0.40D_{DTI} + 0.60D_{DSTI}.$$

Therefore, the effective indicator-level weights are:

Indicator	Effective Weight
Savings Strength	25%
DTI	10%
DSTI	15%
Emergency Coverage	25%
Expense Stability	25%
Total	100%

5.1 Why 25:25:25:25?

The equal weighting was not derived from an assumption that every dimension is exactly equally important in every household. It was selected as a transparent baseline because the Phase I dataset is synthetic and contains no independent observed resilience outcome from which statistically optimal weights could be estimated.

The weighting process therefore followed three principles:

1. each dimension represents a distinct mechanism of financial resilience;
2. no dimension should dominate the index without empirical evidence supporting that choice; and
3. the baseline should be simple enough to reproduce and interpret.

A broad sensitivity analysis then tested whether this choice materially changed household rankings. Each dimension was separately assigned weights from 10% to 40%, with the remaining weight redistributed equally across the other three dimensions.

The minimum rank correlation with the baseline was:

0.9707

and the minimum score correlation was:

0.9800

.

The lowest rank correlation occurred when Expense Stability received 10% and the other dimensions received 30% each.

Takeaway: The 25:25:25:25 structure should be understood as a transparent baseline, not as an empirically “correct” weighting. The sensitivity analysis is what provides evidence that the model is not highly dependent on this exact choice.

5.2 Why These Four Dimensions?

The four dimensions were selected because they represent different ways in which a household can become vulnerable.

Savings Strength captures the ability to create a surplus from current income. Debt Sustainability captures obligations that must be met in future periods. Emergency Coverage captures the stock of liquid resources available immediately when normal income or expenditure conditions change. Expense Stability captures the uncertainty in the household’s financial requirements.

This separation matters because a household can perform well on one dimension and poorly on another. A household can have a strong savings rate but little accumulated liquidity. Another can have stable expenditure but a large debt-service burden.

The dimensions are not assumed to be statistically independent. In the simulation, Savings Strength and Emergency Coverage had a correlation of 0.432, while Debt Sustainability and Emergency Coverage had a correlation of 0.764.

The stronger debt-emergency relationship is partly produced by the simulation mechanics: households experiencing deficits use savings first and may borrow after savings are exhausted.

Takeaway: The correlations indicate overlap, not duplication. Each dimension remains because it represents a different financial mechanism even when the mechanisms interact.

6 Phase I: Simulation and Results

The HFRI was implemented in Python using a synthetic population of 1,000 households observed for 12 months, producing:

$$1,000 \times 12 = 12,000$$

household-month observations.

The simulation generated variation in income, expenditure, expenditure volatility, essential expenditure, liquid savings, debt, interest rates, debt maturities and debt service. A fixed random seed of 42 was used for reproducibility.

The monthly financial process was designed so that:

- debt service and expenditure were deducted from income;
- monthly surpluses increased liquid savings;
- deficits first reduced available savings;
- remaining deficits could generate new borrowing subject to a six-month baseline-income borrowing cap; and
- any remaining deficit was recorded as unmet.

Monthly income was generated around a median of approximately Rs. 24,000, with a range of Rs. 6,000 to Rs. 150,000. The expenditure ratio varied between 0.55 and 1.25. Approximately 35% of households initially had debt. Existing debt was assigned different interest rates and maturities, while new borrowing was given higher interest rates and shorter terms.

These assumptions create financially distinct households. They are simulation assumptions rather than estimates of the actual distribution of Indian households.

The final simulation produced:

Statistic	HFRI
Mean	75.839
Median	82.043
Standard deviation	16.245
Minimum	16.270
10th percentile	48.402
25th percentile	67.569
75th percentile	87.687
90th percentile	91.354
95th percentile	92.966
Maximum	97.235

The component means were:

Component	Mean Score
Savings Strength	78.29
Debt Sustainability	88.22
Emergency Coverage	62.35
Expense Stability	74.50

Emergency Coverage had the lowest mean because a meaningful share of the simulated households held limited liquid savings relative to essential expenditure.

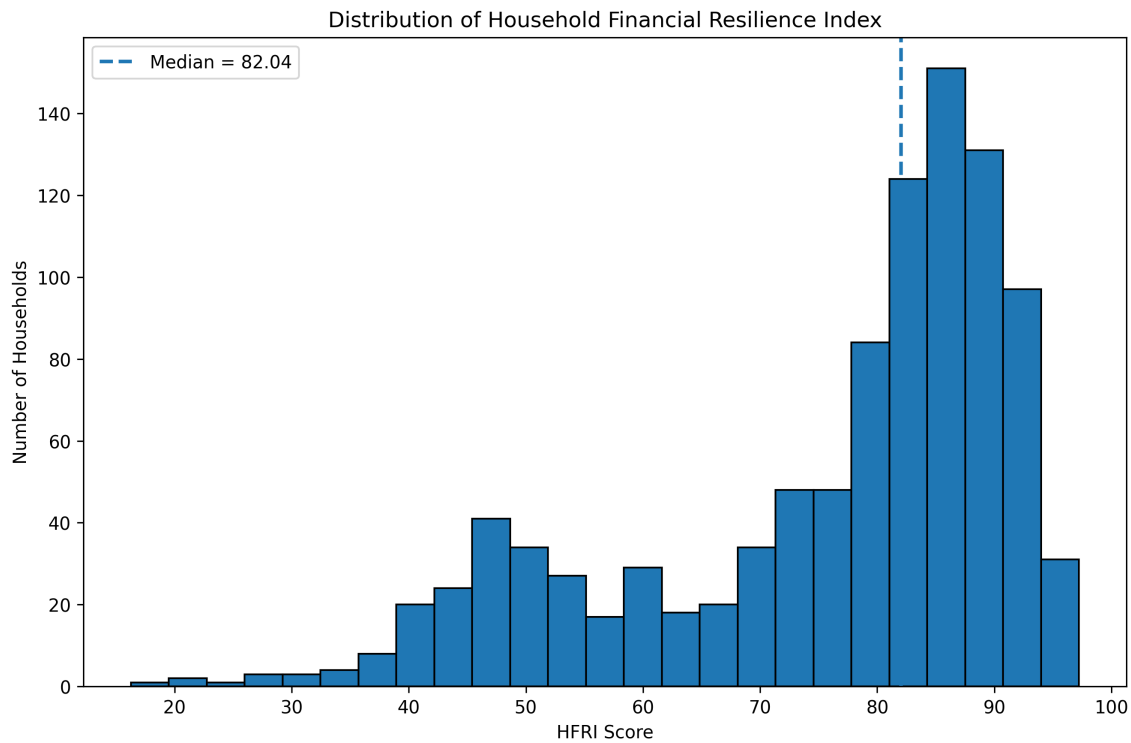


Figure 1: *Distribution of HFRI scores across the 1,000 simulated households.*

A useful diagnostic was the group of 58 households with HFRI scores between 45 and 50. Their mean Emergency Coverage score was only 0.15, compared with 62.35 for the full sample, while their mean Debt Sustainability score was 37.91 versus 88.22 for the full sample. Their Savings Strength and Expense Stability scores were much closer to the overall means.

This concentration therefore reflects a combination of depleted liquidity and weak debt sustainability rather than a mathematical discontinuity.

Takeaway: HFRI successfully separates households with substantially different financial positions within the simulated environment. The low-score region is not produced by a single formula breakpoint; it is produced by combinations of weak financial conditions.

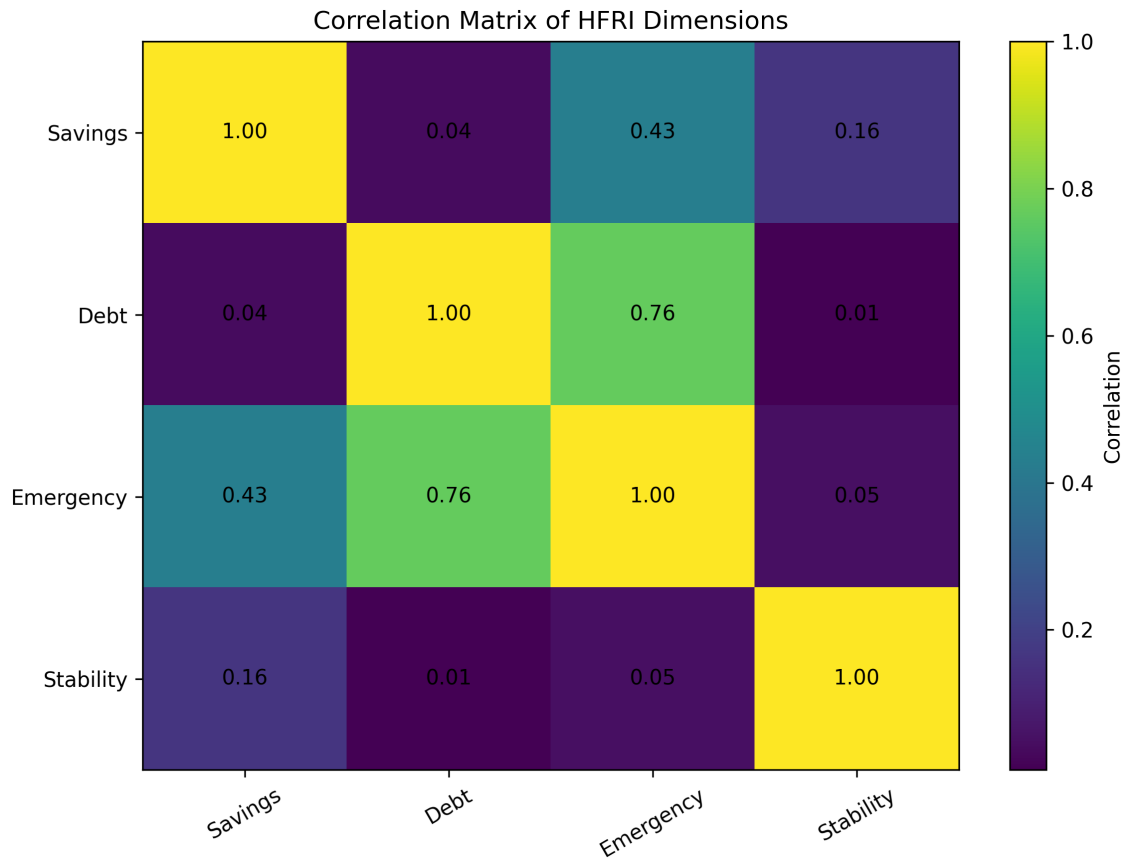


Figure 2: Correlation matrix of the four HFRI dimensions across the simulated households.

The component correlations were:

	Savings	Debt	Emergency	Stability
Savings	1.000	0.037	0.432	0.161
Debt	0.037	1.000	0.764	0.010
Emergency	0.432	0.764	1.000	0.052
Stability	0.161	0.010	0.052	1.000

The HFRI also moved in the expected direction relative to its underlying indicators:

Indicator	Correlation with HFRI
Savings Rate	+0.4805
DTI	-0.6636
DSTI	-0.5854
Emergency Coverage	+0.7332
Expense CV	-0.3009

These relationships provide evidence of internal computational coherence.

Takeaway: Higher savings and liquidity are associated with higher HFRI scores, while higher debt burdens and expenditure volatility are associated with lower scores. These are internal relationships within the simulation, not causal or population-level estimates.

7 Phase II: External Benchmark and Shock Testing

Phase II extends the model in two directions. First, the HFRI is compared with an established real-world financial-resilience framework. Second, the simulated households are exposed to defined income and

expenditure shocks.

The purpose is not to claim that HFRI has been externally validated. Instead, Phase II asks whether the index is consistent with an established benchmark and whether it behaves coherently when simulated households experience financial stress.

7.1 Why the ONS Benchmark?

The Office for National Statistics was selected because its financial-resilience measure is transparent, publicly documented, and directly related to the ability of households to absorb an income shock.

The ONS defines financial resilience partly through the ability to cope with sudden falls in income or unavoidable rises in expenditure. Its experimental measure asks whether formal financial assets are sufficient to cover a specified fall in employment income for three months [27].

For a 25% income reduction, the ONS compared household financial assets with:

$$0.25 \times \text{monthly income} \times 3.$$

The ONS analysis used approximately 18,000 households from Round 6 of the Wealth and Assets Survey, covering April 2016 to March 2018 [27].

The benchmark was selected for four reasons:

1. it directly measures financial shock absorption;
2. its methodology is transparent;
3. its 25%-for-three-months structure can be reproduced in the HFRI simulation; and
4. it does not require a proprietary scoring system.

The ONS measure is not identical to HFRI. It includes a broader range of formal financial assets than the liquid savings variable in the simulation. Therefore, liquid savings were used as the closest comparable measure.

7.2 ONS-Style Benchmark

For each simulated household:

$$B_{ONS} = 0.25 \times Y_m \times 3$$

where Y_m is average monthly income.

A household was classified as ONS-style resilient if:

$$L \geq B_{ONS}.$$

The resulting benchmark rate was:

$$75.9\%$$

The relationship with HFRI was:

$$r_{\text{Pearson}} = 0.8729$$

and:

$$\rho_{\text{Spearman}} = 0.7339.$$

ONS-style classification	n	Mean HFRI	Median HFRI
Not resilient	241	50.69	50.08
Resilient	759	83.83	84.79

Across HFRI quartiles:

HFRI Quartile	Mean HFRI	ONS-style survival
Q1	50.84	7.6%
Q2	76.57	96.0%
Q3	84.90	100.0%
Q4	91.05	100.0%

Takeaway: The ONS-style benchmark produces a strong separation between low- and high-HFRI households within the simulation. This supports consistency with the benchmark, but it should not be called independent external validation because the HFRI and benchmark both depend substantially on liquid savings.

7.3 Income Shock

Each household was subjected to a 25% reduction in monthly income for three months.

The total income loss was:

$$Loss = 0.25 \times Y_m \times 3.$$

Final liquid savings were used first. If savings were insufficient, the remaining amount was recorded as a borrowing requirement.

The overall results were:

Measure	Result
Shock	-25% income for 3 months
Savings survival	75.9%
Borrowing required	24.1%
Mean borrowing requirement	Rs. 4,236

Results by HFRI quartile were:

Quartile	Mean HFRI	Survival	Mean Borrowing Need
Q1	50.84	7.6%	Rs. 16,841
Q2	76.57	96.0%	Rs. 103
Q3	84.90	100.0%	Rs. 0
Q4	91.05	100.0%	Rs. 0

The 75.9% survival rate is numerically identical to the ONS-style benchmark because the same 25%-for-three-months liquidity criterion is used.

Takeaway: The income shock produces a very strong gradient across HFRI quartiles. Almost all households above the first quartile survive the specified shock using existing liquidity, while the lowest quartile has substantial borrowing requirements.

Important qualification: The income-shock result and the ONS-style benchmark are not two independent tests. They are two presentations of the same liquidity criterion.

7.4 Expense Shock

The second stress test increased monthly expenditure by 25% for three months:

$$\text{Additional Expenditure} = 0.25 \times C_m \times 3.$$

This is a researcher-defined stress scenario rather than an ONS benchmark.

The overall results were:

Measure	Result
Shock	+25% expenditure for 3 months
Savings survival	76.6%
Borrowing required	23.4%
Mean borrowing requirement	Rs. 3,372

Results by HFRI quartile were:

Quartile	Mean HFRI	Survival	Mean Borrowing Need
Q1	50.84	8.8%	Rs. 13,445
Q2	76.57	97.6%	Rs. 43
Q3	84.90	100.0%	Rs. 0
Q4	91.05	100.0%	Rs. 0

Takeaway: The expense shock produces a similar pattern to the income shock. Households in the lowest HFRI quartile are much more likely to require additional borrowing, while the two highest quartiles absorb the shock using existing liquidity.

7.5 HFRI Response to Shocks

The stress-state HFRI was recalculated after reducing available liquid savings while keeping the other baseline financial characteristics unchanged.

Scenario	Mean HFRI	Median HFRI	Mean Change
Baseline	75.839	82.043	–
Income shock	73.647	79.218	-2.191
Expense shock	74.042	79.912	-1.797

By HFRI quartile, the income shock produced mean changes of:

$$-1.341, \quad -4.322, \quad -2.088, \quad -1.014$$

for Q1 through Q4 respectively.

For the expense shock, the corresponding changes were:

$$-1.253, \quad -3.633, \quad -1.600, \quad -0.703.$$

The stress-state calculation is exploratory. It demonstrates that the index responds when liquidity deteriorates, but it is not a separately validated version of HFRI.

Takeaway: HFRI declines under both financial shocks, as expected. The change is not uniform across households because the same liquidity loss has different consequences depending on the household's existing financial position.

8 A Model-Based Interpretation Scale

A continuous 0–100 index allows households to be compared, but users may still want to understand what different scores imply.

The interpretation bands were therefore not created first and then imposed on the data. Instead, they were derived after the simulation by examining how shock survival changed across HFRI score ranges.

HFRI Band	Households	Income Survival	Expense Survival
< 60	192	0–10%	0–10%
60–65	35	22.9%	25.7%
65–70	34	55.9%	67.7%
70–75	71	94.4%	95.8%
75–100	668	98.9–100%	100%

The most important transition occurs around 65–70.

At 60–65, fewer than one-quarter of households survived either shock.

At 65–70, survival increased to 55.9% for the income shock and 67.7% for the expense shock.

At 70–75, survival increased again to above 94% for both scenarios.

The resulting provisional interpretation is:

HFRI Score	Model-Based Interpretation
< 60	Very low simulated shock-absorption capacity
60–65	Low simulated shock-absorption capacity
65–70	Transition zone
70–75	High simulated shock-absorption capacity
75–100	Very high simulated shock-absorption capacity

The justification for the bands is therefore empirical *within the model*: the bands correspond to changes in observed simulated shock-survival rates rather than arbitrary round-number labels.

However, a score of 70 is not a universal real-world financial-resilience threshold. It is the point around which the present simulation shows a sharp increase in the ability to absorb the two specified shocks.

The interpretation scale also has a structural limitation. Shock survival depends heavily on liquid savings, and liquid savings already form one of the four HFRI dimensions. A strong relationship between HFRI and shock survival is therefore partly expected from the construction of the index.

Takeaway: The 65–70 transition is useful as a provisional interpretation of this model, but the continuous HFRI score remains the primary output. The bands should only become population-level classifications after testing against real household outcomes.

9 Discussion

The results answer the research question at the level of the present model. HFRI can provide a transparent and multidimensional measure that combines several financial mechanisms into a common scale.

The framework has several strengths. First, the underlying variables are simple and interpretable. Second, the scoring functions are bounded and directionally consistent. Third, the weighting structure is explicit and reproducible. Fourth, the sensitivity analysis shows that household ordering is not highly dependent on moderate changes in dimension weights. Finally, Phase II shows that higher HFRI values are associated with substantially greater simulated capacity to absorb defined financial shocks.

The results also identify what the model does not establish. The synthetic population is not representative of Indian households. The score anchors are researcher-defined. The equal dimension weights are a modelling choice. Expense Stability is more exploratory than savings, debt and liquidity. The ONS comparison is a benchmark rather than independent real-world validation.

The distinction between internal coherence and empirical validity is central. A high R^2 during calibration means that a selected mathematical function reproduces the chosen calibration points. A high correlation between HFRI and the ONS-style benchmark means that the index is consistent with that benchmark within the simulation. Neither result proves that a particular HFRI score corresponds to a particular probability of real-world financial distress.

The strongest next step is therefore empirical testing. HFRI should be calculated using real household-level financial data where income, expenditure, debt, liquid savings and expenditure variation can be observed. Scores can then be compared with independent outcomes such as borrowing, savings depletion, missed payments, financial shortfalls, or the ability to meet unexpected expenses.

10 Limitations

Eight main limitations remain.

- 1. Synthetic rather than observed households.** The computational population is simulated rather than drawn from an observed representative household dataset. The results therefore demonstrate how HFRI behaves under the specified assumptions rather than describing the actual distribution of resilience in any population.
 - 2. Researcher-defined calibration points.** The literature provides meaningful financial landmarks but does not provide a universal mapping from savings rates, debt ratios, emergency coverage or expenditure CV into a 0–100 resilience score. The numerical score anchors are therefore modelling assumptions.
 - 3. Researcher-defined weighting.** The 40:60 DTI-DSTI weighting and 25:25:25:25 four-dimensional weighting are not estimated from an independent outcome. Sensitivity analysis demonstrates robustness, but not statistical optimality.
 - 4. Simplified household behaviour.** Real households may reduce discretionary spending, sell assets, refinance debt, receive transfers, change employment, delay payments or obtain family support following a shock. These responses are simplified in the simulation.
 - 5. No independent observed resilience outcome.** The simulation does not contain real-world outcomes such as default, arrears, bankruptcy, forced asset sales, missed essential payments or observed consumption reductions following a real financial shock. Predictive validity therefore cannot be established.
 - 6. ONS is not identical to HFRI.** The ONS measure uses broader formal financial assets and real household survey data, whereas the HFRI comparison uses liquid savings within a synthetic population. The comparison is therefore benchmarking rather than validation.
 - 7. Expense Stability remains exploratory.** The relationship between expenditure volatility and financial resilience is less established as a standalone index component than savings, debt and liquid assets. Higher volatility can sometimes reflect financial flexibility rather than financial weakness.
 - 8. Shock tests are model-specific.** The income and expenditure shocks represent two controlled scenarios. They do not capture the full range of real financial shocks, their durations, household behavioural responses, or interactions with inflation, employment, interest rates and asset prices.
- Overall limitation:** The current evidence establishes mathematical coherence and internal robustness, not population-level accuracy.

11 Conclusion

This study developed the Household Financial Resilience Index as a transparent mathematical framework for measuring a multidimensional concept.

The HFRI combines Savings Strength, Debt Sustainability, Emergency Coverage and Expense Stability:

$$HFRI = 0.25S + 0.25D + 0.25E + 0.25X$$

with:

$$D = 0.40D_{DTI} + 0.60D_{DSTI}.$$

The 1,000-household simulation produced a wide distribution of scores, from 16.27 to 97.24. The index moved in the expected direction relative to its underlying indicators, and the household ordering remained highly stable under substantial changes to the dimension weights.

Phase II provided two additional tests of the framework. First, HFRI was compared with an ONS-style liquidity benchmark based on the ability to absorb a 25% income reduction for three months. Second, the households were exposed to both income and expenditure shocks. Higher HFRI households were substantially more likely to absorb both shocks without additional borrowing.

The model also produced a clear transition around HFRI 65–70. Shock survival increased sharply through this region, exceeding 94% for both scenarios in the 70–75 band. This supports a provisional model-based interpretation scale, but not a universal real-world threshold.

The central conclusion is therefore measured:

HFRI is internally coherent, mathematically transparent and responsive to simulated financial stress, but it is not yet an empirically validated measure of real household financial resilience.

Its next stage should be empirical testing using observed household-level data. If real-world outcomes show systematic differences in financial stress and shock absorption across HFRI scores, the calibration functions, weights and interpretation bands can then be refined on an empirical basis.

Data and Code Availability

The simulation and calculations were implemented in Python using Google Colab. The notebook contains the household simulation, HFRI calculation, correlation analysis, weighting sensitivity analysis, Phase II benchmarking, shock tests and interpretation-scale analysis.

https://colab.research.google.com/drive/1uz_USmIzrsyE7bu4OyV3C2V7j-ltINxL?usp=sharing

A fixed random seed of 42 was used so that the reported computational results can be reproduced from the same model specifications.

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